

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 1/14 - U_MCU

COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware.
Required because RP2040 has no internal flash.
BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference.
Load capacitors and XOUT resistor support
oscillation and limit crystal drive.

U_PD - CH224K

15 V request: CFG1=0, CFG2=CFG3=1.
PG is open-drain, active low. PD-only mode:
chip DP/DM shorted; VBUS sense pin unused.

U_BUCK - AP63203WU-7

Efficient 3.3 V supply for MCU and logic.
Diode OR accepts USB data power or PD.
Inductor and capacitors smooth switching.

U_DRV - A4988

Fixed 1/16 steps; 10k enable/sleep defaults.
3.9k/1k REF divider, 0.24 ohm sense resistors:
0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

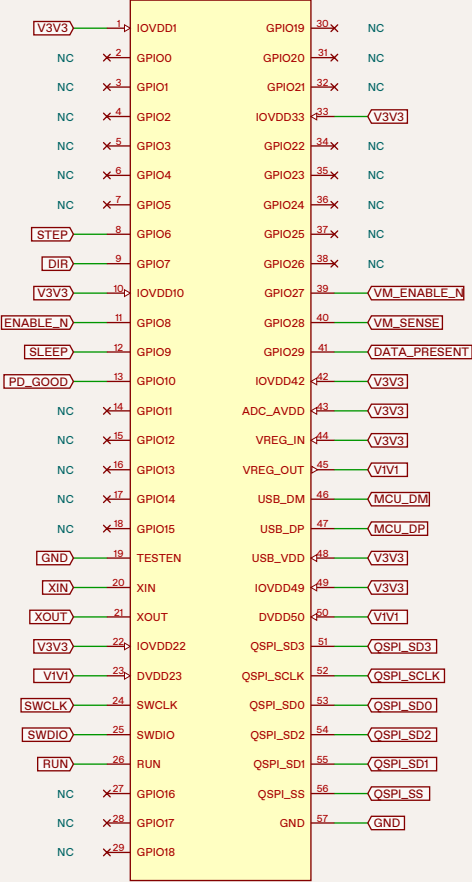
U_VM_ISO - SN74CBTLV1G125

Isolates VM ADC when logic power is off.
10k OE pull-up defaults off; MCU gates ON.
Bus/Z1 when ON; wait 0.5 ms then sample.

Q_DATA - MMBT3904-7-F

Detects host 5 V without direct GPIO feed.
10k base feed and 10k 3.3 V collector pull.
DATA_PRESENT LOW means host attached.

U_MCU



RP2040

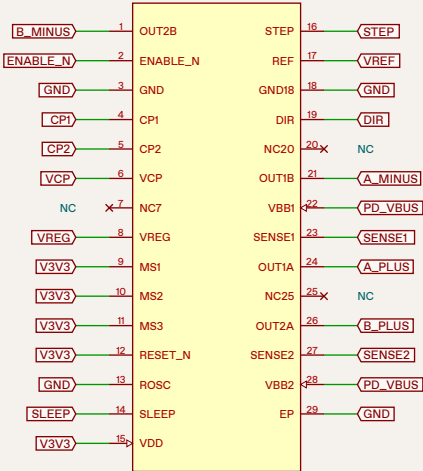
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PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

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SHEET 2/14 - U_DRV

U_DRV



A4988SETTR-T

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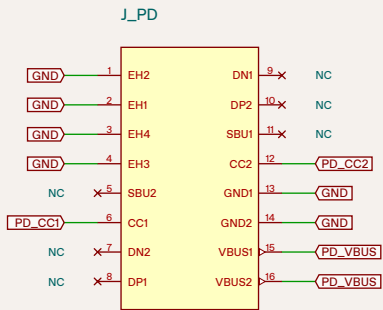
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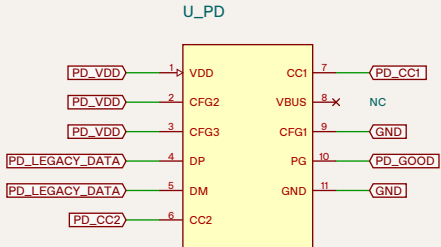
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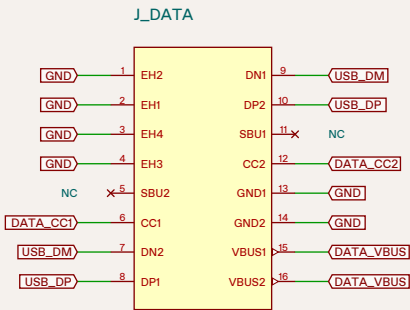
SHEET 3/14 - J_PD, J_DATA, U_PD, R_PD, C_PD, R_PG



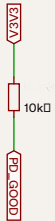
TYPE-C-31-M-12



CH224K



TYPE-C-31-M-12



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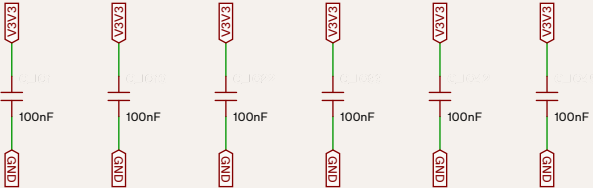
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SHEET 4/14 - C_IO1, C_IO10, C_IO22, C_IO33, C_IO42, C_IO49



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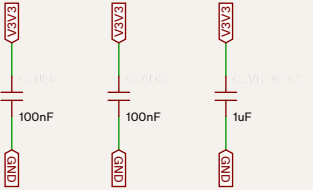
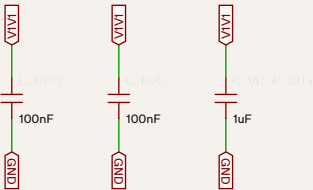
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SHEET 5/14 - C_DV23, C_DV50, C_USB, C_ADC, C_VREG_IN, C_VREG_OUT



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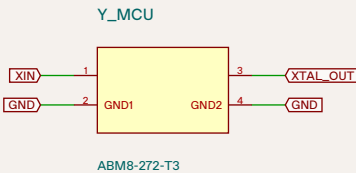
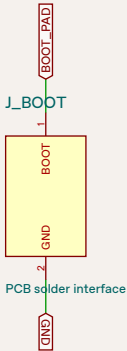
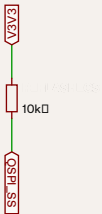
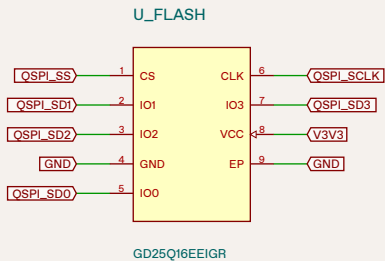
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SHEET 6/14 - U_FLASH, C_FLASH, R_FLASH_CS, R_BOOT, J_BOOT, Y_MCU



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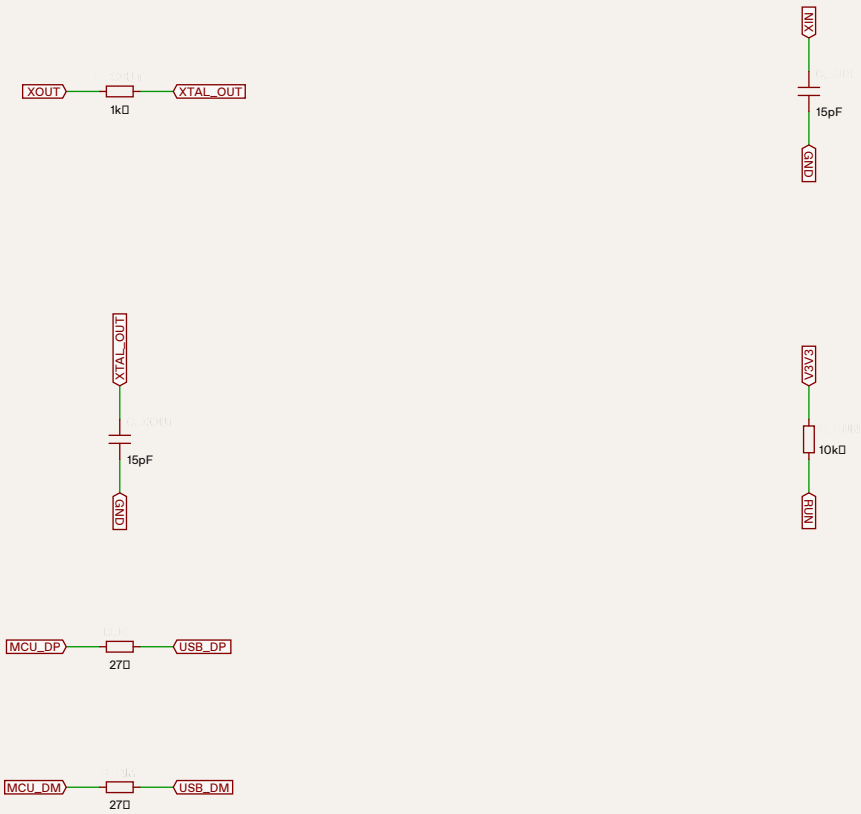
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NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 7/14 - R_XOUT, C_XIN, C_XOUT, R_RUN, R_DM, R_DP



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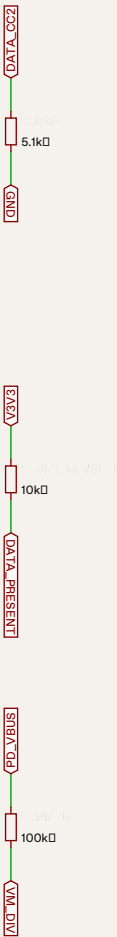
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SHEET 8/14 - R_CC1, R_CC2, R_USB_SENSE_H, R_USB_SENSE_L, Q_DATA, R_VM_H



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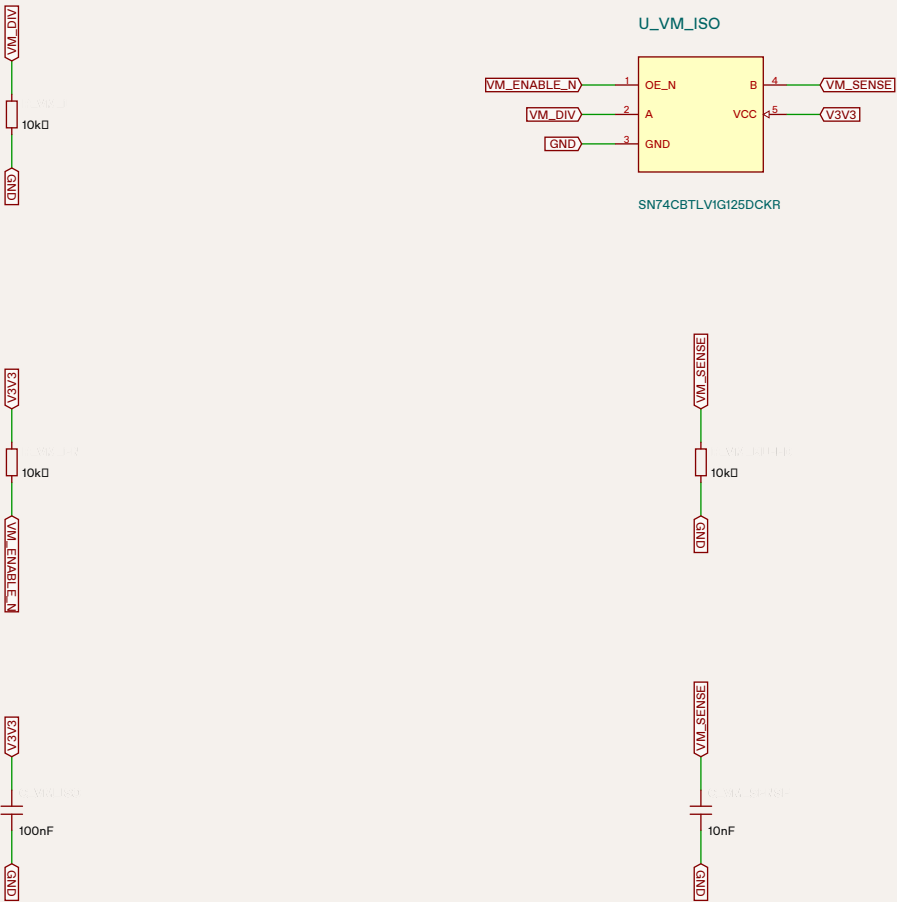
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SHEET 9/14 - R_VM_L, U_VM_ISO, R_VM_EN, R_VM_BLEED, C_VM_ISO, C_VM_SENSE



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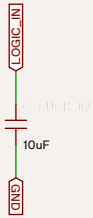
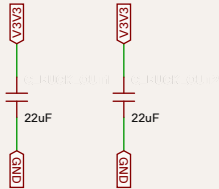
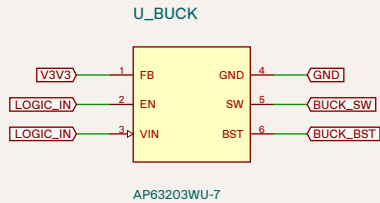
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SHEET 10/14 - U_BUCK, L_BUCK, C_BOOTSTRAP, C_BUCK_IN, C_BUCK_OUT1, C_BUCK_OUT2



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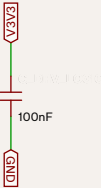
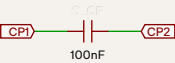
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SHEET 11/14 - D_PD, D_DATA, C_CP, C_VCP, C_VREG, C_DRV_LOGIC



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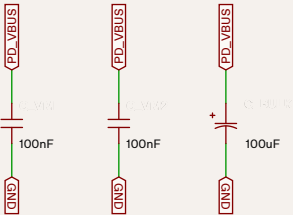
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Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

U_VM_ISO - SN74CBTLV1G125
Isolates VM ADC when logic power is off.
10k OE pull-up defaults off; MCU gates ON.
Bus/Z1 when ON; wait 0.5 ms then sample.

Q_DATA - MMBT3904-7-F
Detects host 5 V without direct GPIO feed.
10k base feed and 10k 3.3 V collector pull.
DATA_PRESENT LOW means host attached.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 12/14 - C_VM, C_VM2, C_BULK, R_SA, R_SB, R_REF_H



COMPONENT PURPOSE

U_MCU - RP2040
USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR
2 MB external QSPI flash stores firmware.
Required because RP2040 has no internal flash.
BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3
12 MHz crystal is the USB clock reference.
Load capacitors and XOUT resistor support
oscillation and limit crystal drive.

U_PD - CH224K
15 V request: CFG1=0, CFG2=CFG3=1.
PG is open-drain, active low. PD-only mode:
chip DP/DM shorted; VBUS sense pin unused.

U_BUCK - AP63203WU-7
Efficient 3.3 V supply for MCU and logic.
Diode OR accepts USB data power or PD.
Inductor and capacitors smooth switching.

U_DRV - A4988
Fixed 1/16 steps; 10k enable/sleep defaults.
3.9k/1k REF divider, 0.24 ohm sense resistors:
0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6
Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

U_VM_ISO - SN74CBTLV1G125
Isolates VM ADC when logic power is off.
10k OE pull-up defaults off; MCU gates ON.
Bus/Z1 when ON; wait 0.5 ms then sample.

Q_DATA - MMBT3904-7-F
Detects host 5 V without direct GPIO feed.
10k base feed and 10k 3.3 V collector pull.
DATA_PRESENT LOW means host attached.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 13/14 - R_REF_L, C_REF, R_ENABLE, R_SLEEP, J_MOTOR, J_DEBUG

COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR. Checks PD voltage before enabling the motor. Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware. Required because RP2040 has no internal flash. BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference. Load capacitors and XOUT resistor support oscillation and limit crystal drive.

U_PD - CH224K

15 V request: CFG1=0, CFG2=CFG3=1. PG is open-drain, active low. PD-only mode: chip DP/DM shorted; VBUS sense pin unused.

U_BUCK - AP63203WU-7

Efficient 3.3 V supply for MCU and logic. Diode OR accepts USB data power or PD. Inductor and capacitors smooth switching.

U_DRV - A4988

Fixed 1/16 steps; 10k enable/sleep defaults. 3.9k/1k REF divider, 0.24 ohm sense resistors: 0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6

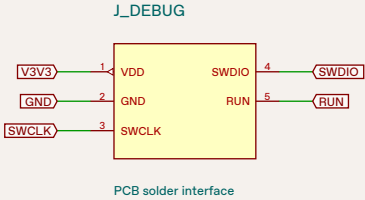
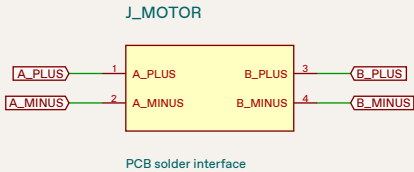
Clamps ESD on USB D+ and D- at the data port. VBUS pin references the protection diodes.

U_VM_ISO - SN74CBTLV1G125

Isolates VM ADC when logic power is off. 10k OE pull-up defaults off; MCU gates ON. Bus/Z1 when ON; wait 0.5 ms then sample.

Q_DATA - MMBT3904-7-F

Detects host 5 V without direct GPIO feed. 10k base feed and 10k 3.3 V collector pull. DATA_PRESENT LOW means host attached.

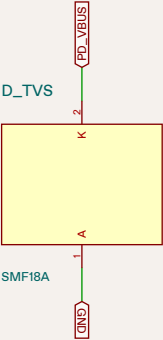
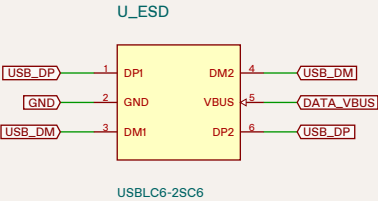


Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_RP2040 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 14/14 - U_ESD, D_TVVS



COMPONENT PURPOSE

U_MCU - RP2040

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal regulator supplies the 1.1 V core.

U_FLASH - GD25Q16EEIGR

2 MB external QSPI flash stores firmware.
Required because RP2040 has no internal flash.
BOOT pads permit ROM USB boot recovery.

Y_MCU - ABM8-272-T3

12 MHz crystal is the USB clock reference.
Load capacitors and XOUT resistor support
oscillation and limit crystal drive.

U_PD - CH224K

15 V request: CFG1=0, CFG2=CFG3=1.
PG is open-drain, active low. PD-only mode:
chip DP/DM shorted; VBUS sense pin unused.

U_BUCK - AP63203WU-7

Efficient 3.3 V supply for MCU and logic.
Diode OR accepts USB data power or PD.
Inductor and capacitors smooth switching.

U_DRV - A4988

Fixed 1/16 steps; 10k enable/sleep defaults.
3.9k/1k REF divider, 0.24 ohm sense resistors:
0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the protection diodes.

U_VM_ISO - SN74CBTLV1G125

Isolates VM ADC when logic power is off.
10k OE pull-up defaults off; MCU gates ON.
Bus/Z1 when ON; wait 0.5 ms then sample.

Q_DATA - MMBT3904-7-F

Detects host 5 V without direct GPIO feed.
10k base feed and 10k 3.3 V collector pull.
DATA_PRESENT LOW means host attached.

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PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.